

gradients near convergence. `tightbounds` only measures bounds separation for non-contradictory states, so the "flip-flop" pattern extends to this loss component.

The pattern is probably just an artifact of the loss function, and the added degrees of freedom for parameter updates gradient-introduction in the clamped Łukasiewicz domain gives. A standard decreasing learning rate, which also benefits from applied gradient clipping, can ensure some learning convergence at a degree of the real optimum.

A conventional Łukasiewicz conjunction `False` gradient $c_{\text{grad}} = 0$ converges slower than with non-zero gradients, as there are fewer parameters with gradients. In practice, we implement disjunctions via De Morgan's law in terms of negations and conjunctions. However, for disjunctive clauses and `True`-polarized axioms, there could still be sufficient gradient-tracked parameters for conventional Łukasiewicz to change disjunctive clauses from `False` to `True` (the clamped region for disjunctions) to remove contradictions in this system of formulae. However, without gradients, one would normally not change a clamped result, e.g. `True` to `False` for disjunctions.

In the case of these CNF formulae, gradient introduction across the full operating domain of logical connectives may not be strictly necessary to remove contradictions. However, for more complex systems of formulae and facts a non-zero gradient in the clamped regions of logical connectives may be required to address some contradictions. Sigmoidal and tailored activation functions can also provide full gradient coverage for logical neurons.

H.2 LUBM benchmark

This section gives additional details on **LUBM benchmark** experiments as reported in **Empirical evaluation** section (Section 7) of the main paper, on pages 7 and 8 (line numbers from 245 to 257). Our main goal in these experiments (using LUBM data) is to demonstrate the soundness and completeness of the reasoning performed by LNN. In addition, we also want to report on preliminary evaluation of our initial exploration towards achieving noise robustness using LNN, i.e., specifically to explore LNN's ability to handle noisy axioms that result in ontological inconsistencies.

Dataset. Lehigh University Benchmark (LUBM) [10] is a synthetic OWL reasoning dataset in university domain together with 14 benchmark queries. In our experiments we considered LUBM data generated for 1 university. This has in total 102707 facts expressed in triples format in addition to axioms expressing university domain ontology in OWL format. We pre-processed these facts and axioms to convert them into equivalent LNN constructs so they could be consumed directly by our LNN code.

Soundness and completeness. The purpose of this experiment is to evaluate the soundness and completeness of the results generated by LNN for 14 LUBM benchmark queries. The 14 queries are originally designed to evaluate the accuracy of symbolic reasoners. Accurate answering of these queries require varying degrees of reasoning capabilities from shallow to deep. Thus, accurate answers for all these queries by LNN should assert its deep reasoning capabilities.

In our approach, we built a single LNN for the entire LUBM data, with logical nodes and inter-connections corresponding to various predicate nodes, logical connectives and their interactions, as expressed through the axioms. This resulted in LNN with 257 nodes. The facts expressed are assigned as initial groundings to the corresponding nodes. We then made this initial network to go through multiple passes of forward and backward inferences, to allow all the nodes to gather their inferred groundings together with bound values, through reasoning as per network structure and initial facts. It took 4 bidirectional passes for the network to converge with respect to the node level inference groundings and the corresponding upper and lower bound values. Then for each query, we attach a small query-specific network to the converged LNN in order to infer its groundings through a single forward pass within that particular query network alone. The final groundings computed at the query nodes correspond to the answers for the queries.

In this experiment, we compared LNN with a few standard symbolic reasoners namely Stardog [20], Virtuoso [23] and Blazegraph [2]. A detailed comparison of the results for each of the 14 benchmark queries is given in Table 9. LNN achieves 100% precision and recall, thus demonstrating its sound and complete reasoning capabilities on LUBM OWL reasoning task. Among symbolic reasoners, Stardog also achieves 100%, while Virtuoso and Blazegraph achieve respective average recall values

of only 72% and 78% indicating their lack of complete reasoning capabilities in case of queries that need deep reasoning.

Table 9: Soundness and Completeness for LUBM 1 university showing output sizes for each query

Query	Virtuoso	Blazegraph	Stardog	LNN
Q1	4	4	4	4
Q2	0	0	0	0
Q3	6	6	6	6
Q4	34	34	34	34
Q5	146	719	719	719
Q6	5916	5916	7790	7790
Q7	59	59	67	67
Q8	5916	5916	7790	7790
Q9	103	103	208	208
Q10	0	0	4	4
Q11	224	224	224	224
Q12	15	0	15	15
Q13	1	1	1	1
Q14	5916	5916	5916	5916
Precision	1.0	1.0	1.0	1.0
Recall	0.72	0.78	1.0	1.0

Noise handling. Noisy/incorrect axioms typically result in inconsistent ontology and hamper the ability of the reasoning systems to perform inference. For example, if an axiom makes an incorrect statement about the domain and range for a specific binary variable, that likely would result in inconsistency if there is conflict with domain and range inferred for that variable from the rest of the axioms. Symbolic reasoners typically fail to answer queries when faced with such ontological inconsistencies. Typically, they end up alerting the presence of inconsistency, prompting for correction before proceeding with the inference.

In this experiment we explore the potential use of LNN to handle such ontological inconsistencies. LNNs perform inference by propagating and aggregating truth value bounds at various nodes across the network as per node specific activation. As a result, ontological inconsistencies reflect themselves as truth value bound contradictions at nodes corresponding to incorrect axioms in the network. This is because evidences for truth value bounds computed through different routes of the network are likely to contradict each other when aggregated at the inconsistent nodes. In our preliminary exploration we evaluated the usefulness of this behavior towards identifying the source of inconsistency and to further perform self-correction through down-weighting those parts of the network causing inconsistency, thus effectively switch them off from taking part in further inference.

To evaluate this, we added an incorrect axiom to the original set of LUBM axioms and built LNN using them. The specific noisy axiom we considered makes domain (concept to which subject in the binary predicate belongs) and range (concept to which the object in the binary predicate belongs) incorrect for one of the binary predicates, i.e., it wrongly asserts domain of the subject variable and range of the object variable. For example, we took one of the binary predicates `degreeFrom` with domain and range as `Person` and `University` respectively, and modified range to `Person`. Note that, in addition to the noisy axiom, we also added an additional (correct) axiom to assert that `University` and `Person` concepts are disjoint. This was added to make sure Ontology becomes inconsistent with conflict arising from noisy axiom.

As explained above, during inference such inconsistency is expected to result in bound value contradictions at node corresponding to the incorrect axiom. Once node with contradictory bound values is identified, we down-weight that node to effectively switch that off from taking part in further inferences. The network is then allowed to converge through multiple passes of forward and backward inference (after down-weighting inconsistent node that caused bound value contradictions). The converged network is then used to answer all the 14 benchmark queries. In our experimental run, we found that the LNN is able to successfully locate and down-weight the inconsistent node, converge further through forward and backward inference passes, and then answer all the 14 queries with 100% precision and recall. We currently experimented with simple noise axioms that do not require

deep reasoning to identify conflicts. As a future work, we plan to build upon this idea to further demonstrate the effectiveness of LNN in identifying and self-correcting various types of ontological inconsistencies.

H.3 Additional information on TPTP benchmark

This section gives additional information on the **TPTP benchmark** used in the **Empirical evaluation** section of the main paper. The TPTP (Thousands of Problems for Theorem Provers) [21, 4] is a comprehensive library for Automated Theorem Proving (ATP) systems, with over 22K problems. It contains problems in 53 different domains, represented in 4 different formalisms. These are First-order Form (FOF), Clausal Normal Form (CNF), typed higher-order form (THF) and typed first-order form (TFF). There are a total of 22686 problems, 17535 (77%) of which have equality. The FOF subset contains 8630 problems spanning 41 domains, 6811 (78%) of which have equality. An example problem file is named CSR001+1.p, with CSR indicating the Common Sense Reasoning domain, 001 the problem number in this domain, + indicating FOF format and then 1 after + signifying the difficulty level of the problem. Instead of a + symbol, THF problems are identified by $\wedge$, CNF by $-$ and TFF by $=$.

The CSR domain we evaluated on has a total of 1030 problems with 937 of these in FOF. The difficulty level of its problems range from 1 to 100 with majority of them below level 7. After filtering out problems with functions and equality, the 25 remaining all had a difficulty level of 1. From these 25 problems, the LNN was able to prove all theorems.